

RANDOM FOREST

1

An example: A Practical Problem

- A fish-packing plant wants to automate the process of sorting incoming fish according to species
- Problem: Identifying species of a fish on a conveyor belt
 - ▣ Species: Sea bass and salmon

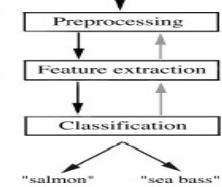
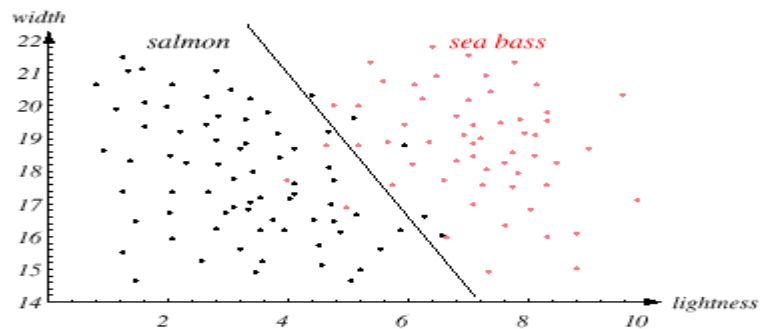


Image source: Pattern Classification by Duda, Hart and Stork

2

Feature Space

- Two features for classification



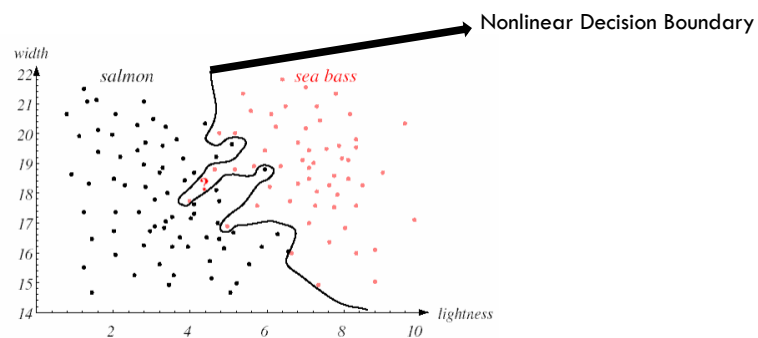
Can we improve the performance further? If yes, how?

Image source: Pattern Classification by Duda, Hart and Stork

3

Feature Space

- Two features for classification



Perfect Classification! Is there a catch?

Image source: Pattern Classification by Duda, Hart and Stork

4

Generalization

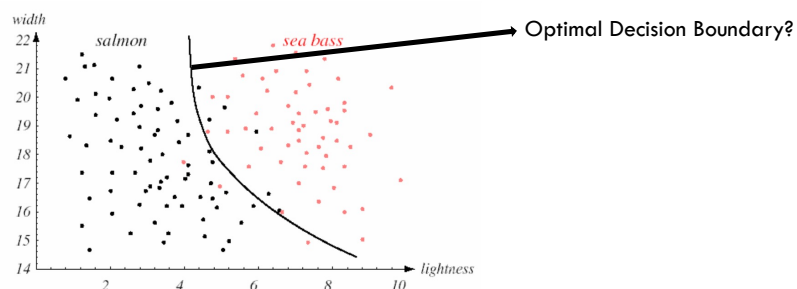
- Classification Goal: Make **accurate predictions** for **new/unseen data** - **Good Generalization**
- The model should NOT be highly tuned to the specific characteristics of the training data – **Overfitting**
- In practice, training data is likely to contain some noise

We are better off with a slightly poorer performance on the training examples if this means that our classifier will have better performance on unseen patterns.

5

Generalization

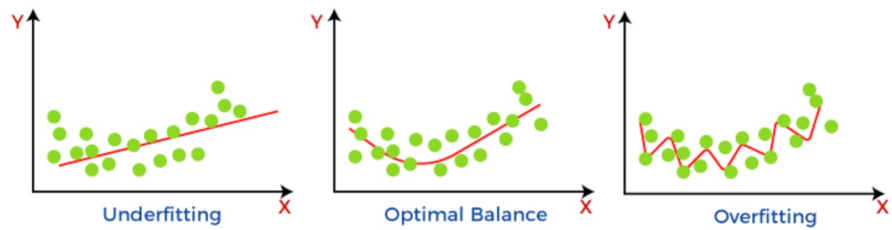
- Classification Goal: Make **accurate predictions** for **new/unseen data** - **Good Generalization**



- A decision boundary that provides an **optimal tradeoff** between **accuracy on the training set** and **unseen data**

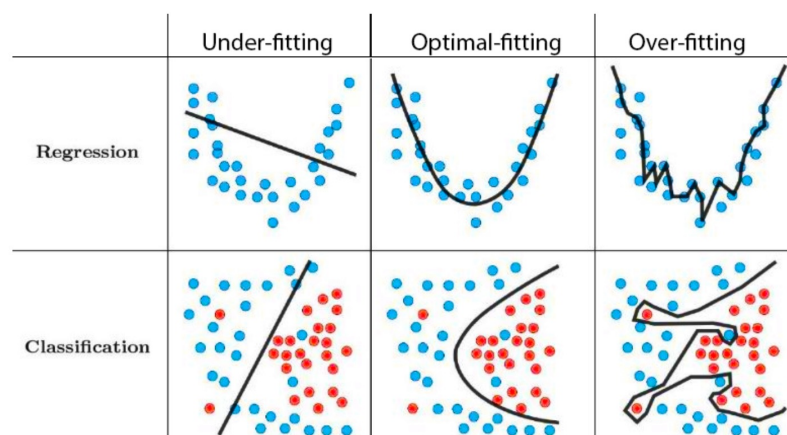
6

Avoid Overfitting and Achieve Optimal Tradeoff



7

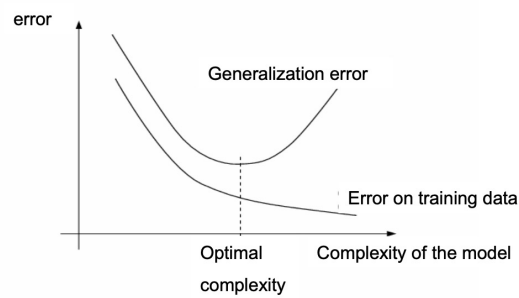
Underfitting and Overfitting



8

Fish Classification Problem: Avoid Overfitting and Achieve Optimal Tradeoff

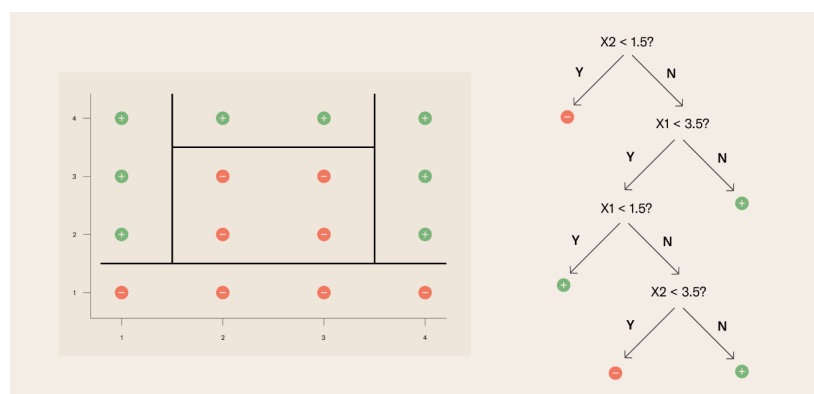
- Evaluate the classifier model on unseen data – **Validation Set**



9

Feature Space

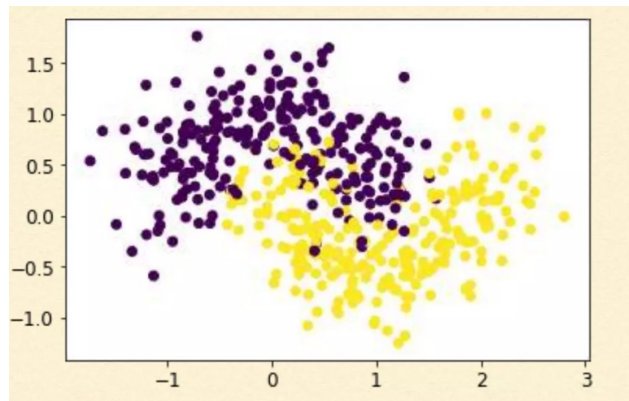
- Decision Boundaries in Decision Tree



12

Avoid Overfitting and Achieve Optimal Tradeoff

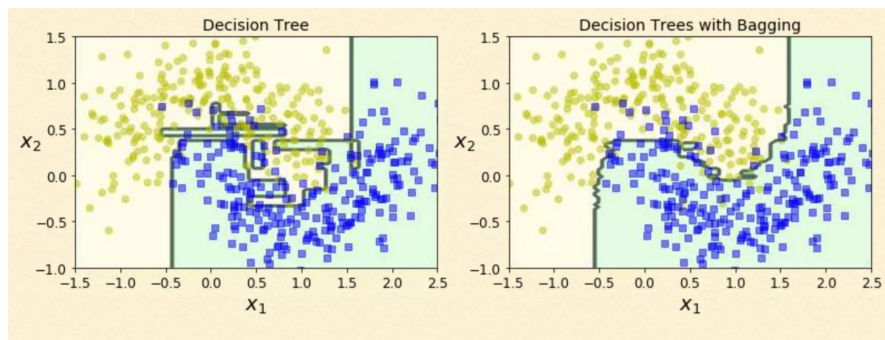
Decision Tree versus Random Forest



13

Avoid Overfitting and Achieve Optimal Tradeoff

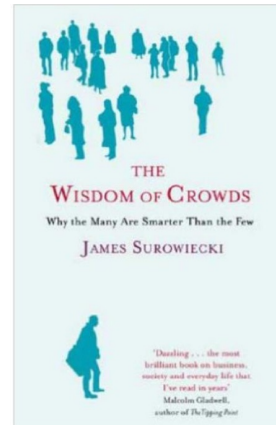
Decision Tree versus Random Forest



14

Random Forest

- **Ensemble learning** is a machine learning technique that aggregates two or more learners to produce better predictions
 - committee-based learning



15

Random Forest

- Base learner, base model, base estimator - refers to the individual models in ensemble algorithms
- consolidating base learner predictions
 - Majority Voting, Averaging

16

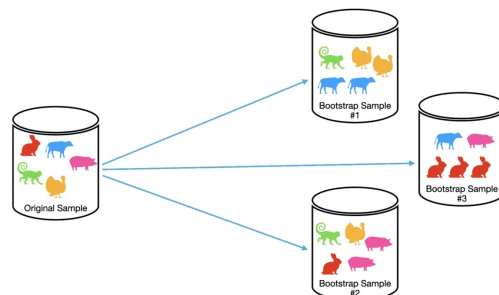
Random Forest

- Random forest uses **bagging** to construct ensembles of **randomized decision trees**
 - Bagging - **bootstrap sampling** and **aggregation**
 - Bootstrap sampling to derive multiple new datasets from one initial training dataset to train multiple base learners

17

Bootstrap Sampling

- **Random** sampling with **replacement**

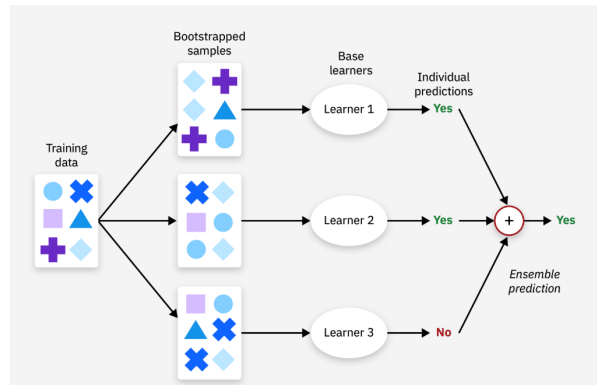


- Each bootstrap sample only contains **approximately 63.2%** of the unique datapoints from the original dataset

18

Random Forest

- Random forest uses **bagging** to construct ensembles of **randomized decision trees**



19

Random Forest

- Random forest uses **bagging** to construct ensembles of **randomized decision trees**
 - considers **random subsets** of features when splitting a node
 - `max_features` parameter
- The **greater diversity** among combined models, the **more accurate** the resulting ensemble model

20

Estimating Generalization Performance: Out-of-bag (OOB) error/score

- **Out-of-bag** samples as unseen data for evaluation
 - Out-of-bag samples are the unique sets of datapoints that are not used for model fitting/training



- Each bootstrap sample only contains **approximately 63.2%** of the unique data points from the original dataset

21

Thank You!

22